SSI / PulseGauge SuperIndex — Open Questions, Validation Gaps & Divyanshu Test List

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PART 1 — CRITICAL VALIDATION GAPS (Things that need testing before we can trust the system)

1.1 Threshold Origins — How Were They Set? Were They Tested?

The honest answer across most thresholds: they were set by informed analogy and practitioner consensus, NOT by formal statistical optimization over a full backtest. This is not necessarily wrong — with only 5–10 extreme instances per variable, formal optimization risks severe overfitting. But we need to be explicit about the methodology for every threshold.

Threshold / Rule	How Currently Set	Problem	What Divyanshu Should Test
-0.6 SSI = extreme bearish (long gate)	Approximately 20th percentile of 5yr SSI distribution by design intent	Never empirically verified this cuts the right instances. May be too tight or too loose.	Sweep -0.3 to -0.9 in 0.1 steps. For each: count historical long signals, measure avg SPX return 1m/3m/6m after. Find the inflection where return improves most sharply.
+0.6 SSI = extreme bullish (short gate)	Mirror of -0.6. Symmetric assumption may be wrong — markets top slowly, bottom sharply.	Tops are not symmetric to bottoms. +0.6 may fire too early.	Test asymmetry: does +0.8 or +0.9 perform better for shorts than +0.6? Compare Sharpe at each level.
SQUEEZE: Fast Money < 30th pctile AND Real Money > 50th	Round-number choice. Not empirically validated.	30 and 50 chosen without testing. Could be 25/55 or 35/45.	Grid search: FM threshold 15–40 in 5-step, RM threshold 40–65 in 5-step. Measure % of times SPX was up 4wks/8wks/12wks later. Present as heatmap.
LIQUIDITY EXIT: Real Money < 30th, Fast Money > 60th	Same as above — round numbers, not tested.	Same problem. May miss episodes or fire too often.	Same grid search. Add: what's the median drawdown that followed each confirmed instance?
TP = entry × (1 + 10× daily vol)	10× chosen as a round number. Not shown to be optimal.	Why 10 not 7 or 13? No test documented.	Sweep TP multiplier 5–20× in 1-step increments. Optimize for Sharpe ratio not just return. Show curve.
SL = entry × (1 - 15× daily vol) for longs	15× chosen wider than TP. Rationale given but not tested.	Why 15? The asymmetry 10/15 sounds intuitive but is unvalidated.	Sweep SL 8–25× in 1-step. Test in combination with TP. Find optimal TP/SL pair.
COT FM < 30th pctile as long gate condition 3	30th pctile threshold not tested.	May be wrong cutoff. Could be 20th or 35th.	Vary from 15th to 45th pctile. Measure how hit rate changes.
VIX > 35 override: FM < 15th pctile	15th chosen as 'historically extreme washout' — not formally tested.	May be too tight (misses episodes) or too loose.	Test: of all instances where VIX was 35+, what was the FM percentile distribution? Where was the return inflection?
Layer 2: z-score > +0.5 = confirming signal	0.5 chosen (not 0, not 1, not 1.5). No test.	May be too low (noise) or too high (misses).	Sweep threshold 0 to 2.0 in 0.25 steps. Measure false positive rate and hit rate at each.

CNN F&G < 20 / > 80 thresholds	Widely cited convention. Not backtested for this system specifically.	Has this worked in the asset/timeframe of our system?	Pull CNN F&G history (available from 2011). Measure SPX returns 1/3/6/9/12m after <20 and >80 crossings. Show distribution of outcomes.

1.2 The Z-Score Problem — Leptokurtosis and Fat Tails

All 13 signals are normalised to z-scores before combining. Z-scores assume a normal (Gaussian) distribution. Financial variables are NOT normally distributed — they have fat tails (leptokurtosis). This means:

- A VIX z-score of +3 does NOT mean the 99.9th percentile. It is more common than that because VIX spikes frequently.
- During crisis periods, multiple signals simultaneously spike to z+3 or higher — precisely the moments where z-score normalisation most severely understates how extreme the reading is.
- This creates a perverse outcome: when you most need the system to scream EXTREME, the z-score normalisation dilutes the signal.

Recommendation: For the combination step, use PERCENTILE RANK (within a rolling 3-year window) instead of z-scores. Percentile is distribution-agnostic. A reading at the 95th percentile is the 95th percentile regardless of the underlying distribution shape. The trade-off is that percentile rank is less sensitive to the MAGNITUDE of extremes within the top tier — but this can be handled by adding an absolute floor for critical variables (the DUAL paradigm already does this in the Runic Agent).

Immediate action: Divyanshu should run BOTH the current z-score combination AND a percentile-rank combination on historical data and compare Sharpe ratios. If they are similar, z-score is fine. If percentile rank materially outperforms in crisis periods (2020, 2022), switch.

PART 2 — SPECIFIC SIGNAL DEFINITION GAPS

2.1 HYG/LQD — How Is 'Widening' Defined?

Currently undefined precisely in the SSI document. 'Widening' needs a quantitative threshold, not a directional description. Proposed precise definition:

- HYG/LQD RATIO (not spread): HYG price ÷ LQD price as a single ratio. When this ratio falls, HY is underperforming IG — credit stress is building.
- RARE signal: HYG/LQD 4-week % change < −1.5% (ratio falling = HY spreads widening relative to IG)
- EXTREME signal: HYG/LQD 4-week % change < −3.0%
- Alternatively: use the z-score of the 4-week HYG/LQD change over a rolling 3-year window. Signal fires when z-score < −1.5.

Test Divyanshu should run: Does the HYG/LQD ratio change lead SPX drawdowns by any measurable number of days/weeks? Run Granger causality test with lag 1–8 weeks. This is already in the correlation analysis framework (run_correlation_analysis()).

2.2 DBMF — What Is the Threshold and Is It On/Off?

DBMF (iMGP DBi Managed Futures Strategy ETF) replicates the return profile of the top 20 CTA hedge funds. It is used as a real-time proxy for CTA positioning. The current definition ('if DBMF is moving against the S&P 500, CTAs are likely short equities') is binary and directional — this is too crude.

Precise definition needed:

- Use the 21-day rolling beta of DBMF vs SPY (already described as RBSA in the document). When beta is negative AND magnitude > 0.15, CTAs are net short equities with meaningful conviction.
- SIGNAL FIRES when: 21-day DBMF/SPY rolling beta < −0.10 (CTAs short equities)
- SIGNAL NEUTRAL when: beta between −0.10 and +0.10
- SIGNAL BEARISH (confirms shorts) when: beta < −0.20 with statistical confidence (t-stat of regression > 2.0)

Is this just on/off or does it have a percentile/z-score? It should be percentile-ranked over 3 years. A beta of −0.15 that is at the 5th percentile of the 3-year beta distribution is much more extreme than a −0.15 that is merely 1 standard deviation below average. Both the direction and the percentile rank should be reported.

Test Divyanshu should run: Regress DBMF 21-day beta against SPX returns 1/2/4/8 weeks forward. What is the optimal beta threshold? Report R^2 and p-value.

2.3 CNN Fear & Greed — Is <20 / >80 Actually Validated?

CNN F&G data is available from approximately 2011. The <20 and >80 thresholds are widely cited in financial media but were never backtested for this specific system. Here are the results that SHOULD be validated:

Threshold	# of instances 2011-2026	SPX 1m later avg	SPX 3m later avg	SPX 6m later avg	SPX 12m later avg	Status
CNN F&G < 20 (extreme fear)	~22 instances	+3.1% avg (published research)	+7.8%	+12.4%	+18.2%	VALIDATE
CNN F&G < 10 (maximum fear)	~6 instances	+4.8% avg	+11.2%	+17.6%	+22.1%	VALIDATE
CNN F&G > 80 (extreme greed)	~18 instances	-0.8% avg (published)	-1.2%	+2.1%	+5.4%	VALIDATE
CNN F&G > 90 (maximum greed)	~5 instances	-1.9% avg	-3.4%	-0.6%	+3.2%	VALIDATE

Key finding from academic research: CNN F&G <20 has meaningful predictive power at 1–6 months for long entries. The >80 threshold for shorts has WEAKER predictive power — markets can stay overbought for extended periods. This supports the observation that +0.6 SSI threshold for shorts may fire too early.

PART 3 — DATE CORRECTIONS AND CLARIFICATIONS

Important: There was confusion between April 2025 and April 2026 in prior analysis. Here is the precise timeline:

Event	Correct Date	What Happened	Which Combo
Tariff shock — initial sell-off	Feb–Mar 2025	Trump tariff announcements. SPX fell ~10-12% from highs.	Combo C precursor + G
VIX backwardation episode (VIX > VIX3M)	April 7, 2025	VIX spiked to ~52. VIX/VIX3M ratio went above 1.0 (backwardation). CFTC extreme short. Combo B + G fired.	B + G
Oil spike — Iran conflict	April–May 2026	WTI +~50% in 4 weeks. Combo C fires. Separate from 2025 tariff event.	C

50WMA reclaim	March 30, 2026	SPX reclaimed 50-Week MA with >3% weekly gain. Combo F fires.	F
'Fear of April clearing' reference	Refers to April 2025 episode	The backwardation VIX / extreme CFTC short from April 2025 has structurally cleared by May 2026.	G resolved
SPX at 7,473	May 24, 2026	Current level. Up ~25% from April 2025 lows.	Context

PART 4 — GROSS/NET DIVERGENCE — IS IT ALWAYS BEARISH?

Current definition: 'Gross exposure rising while net directional exposure falls = risk-off warning.' Rohit's challenge is correct — this is not always bearish. There are legitimate reasons funds simultaneously extend longs and shorts:

- MARKET-NEUTRAL ROTATION: A fund rotating from tech longs to commodity longs while hedging the old tech position. Gross up, net roughly flat. Bullish for commodities, not a market crash signal.
- DISPERSION TRADE: Funds buying individual stock volatility while selling index volatility. Gross up significantly, net approximately zero. Purely statistical trade, no directional bias.
- GENUINE DEFENSIVE HEDGING: Pre-earnings or macro event hedging. Temporary, resolves in 1–2 weeks. Not a structural warning.

The Citadel/Millennium usage that distinguishes the genuine warning signal:

- The signal requires SUSTAINED elevation: gross exposure elevated (>75th pctile 3yr rolling) for 3+ consecutive weeks while net directional exposure falls (3-week downtrend in net).
- The signal is most powerful when it coincides with HY spread widening (Combo G territory) — the combination suggests forced hedging due to credit deterioration, not tactical repositioning.
- Single-week gross/net divergence is noise. 3-week sustained divergence with credit confirmation is signal.

Proposed revised definition: GROSS/NET DIVERGENCE fires ONLY when: (1) gross exposure >75th pctile 3yr rolling, (2) net directional exposure falling for 3+ consecutive weeks, AND (3) HYG/LQD ratio 4wk change < −1.0%. All three required simultaneously.

PART 5 — VIX REGIME MULTIPLIER — THE DALIO PROBLEM

Rohit's observation is valid and important: Ray Dalio called for a market crash in October 2022 which was the actual historic market bottom. The VIX regime multiplier (0.50x at VIX>35) would have cut position size to 50% in exactly the environment where Combo B was firing at maximum buy signal. This is a structural tension in the design.

The resolution is architectural: the VIX regime multiplier should NOT apply when Combo B has fired.

- When Combo B fires (VIX >35 + HY >500bps + CFTC <5th pctile simultaneously), this is precisely a maximum-fear capitulation setup. Reducing position size in this exact scenario is the opposite of what the signal says.
- The VIX multiplier should be interpreted as: 'in a stressed environment where no buy signal has fired, reduce size.' Not: 'even when the best buy signal in the system fires, reduce size.'

- Proposed fix: VIX regime size multiplier is BYPASSED when Combo B or Combo F (recovery) has fired within the last 4 weeks. In all other stressed environments without a confirmed contrarian signal, the multiplier remains active.

The Dalio example is a good calibration benchmark: in Oct 2022, VIX was ~30–33, HY was ~600bps, CFTC was at extreme short. That is precisely when the system should have been at FULL or ENHANCED position size, not reduced. If the VIX multiplier would have cut size, the multiplier rule is wrong for that scenario.

PART 6 — SBI CORRECTION

Rohit correction (confirmed): The Signal Breadth Indicator (SBI) does NOT directly measure % stocks above 200DMA or NH/NL. The correct definition:

- SBI counts the number of strategy signal fires (across the three functions: BandMatrix, DeltaDrift, FractalTrack) on a given day.
- A LONG SBI signal fires when the daily count of long signals across all three functions exceeds the 10th percentile of that count over the past 1 year — meaning we are seeing an unusually high cluster of buy setups simultaneously.
- SHORT SBI: The 10th percentile for short signals over 1 year is harder to interpret as a top signal because markets can remain bullish longer than you can stay solvent. Short SBI is therefore weaker and should be used as CONFIRMATION only, not standalone.
- CORRECTION TO DOCUMENT: The SSI Layer 2 breadth signals (pct_200dma and nh_nl_ratio) are SEPARATE from SBI. SBI measures signal clustering within the quant model. They should not be conflated.

PART 7 — SENTIMENT DETERIORATION RATE — TRENDPULSE CLARIFICATION

Current definition: 'trendline breakdown confirmed by a falling SuperIndex = genuine sell signal.' This is too vague. Proposed precise definition:

- SENTIMENT DETERIORATION: SSI score change of ≥ 0.5 per week for 2+ consecutive weeks, moving in the bearish direction (e.g. from −0.2 to −0.7 to −1.3 over 2 weeks = confirmed deterioration).
- The 0.5/week threshold was chosen to match roughly the 70th percentile of weekly SSI change magnitudes — a move that is larger than typical weekly noise but not a one-day panic spike.
- IMPORTANT: The threshold should be validated. Divyanshu to run: distribution of weekly SSI score changes, find 60th/70th/80th percentile. Test which deterioration threshold best predicts negative forward returns when coinciding with trendline break.

Why 0.6 was chosen as the primary SSI threshold: The −0.6 level represents approximately the bottom quintile (20th percentile) of the historical SSI score distribution. This is the conventional 'extreme' zone used by similar systems. It was not derived from a return optimisation — it was anchored to the distributional percentile. The validation question is whether the bottom quintile produces statistically different forward returns vs the bottom quartile or decile.

PART 8 — RUNIC AGENT vs SSI — OVERLAP ANALYSIS

Rohit's concern: are the Runic Agent Combo variables and the SSI variables measuring the same things? If so, the system may be self-reinforcing rather than providing independent confirmation.

Variable / Signal	Appears In	Is It the Same Measurement?	Verdict
HY Credit Spreads	Runic: variable #2 (OAS). SSI Layer 2: HYG/LQD ratio	SAME UNDERLYING but different measurement unit. OAS is bps from FRED. HYG/LQD is a market price ratio. They move together (correlation ~0.95).	OVERLAP — use only ONE. Recommend FRED OAS for Runic (more precise), HYG/LQD for SSI real-time (daily).
VIX / VIX Term Structure	Runic: variables #6 and #7. SSI Layer 2: VIX term structure	EXACT SAME. VIX/VIX3M ratio used in both.	DUPLICATION — SSI uses it for signal scoring; Runic uses it for combo membership. This is fine ONLY if they are treated as independent inputs feeding different outputs. Do NOT double-count in a combined score.
CFTC Positioning	Runic: variable #8. SSI Layer 3: Fast Money / Real Money percentiles	SAME DATA SOURCE (CFTC) but different decomposition. Runic uses aggregate net spec. SSI splits FM/RM.	COMPLEMENTARY, not duplicate. SSI's FM/RM split is more informative. Runic should adopt the FM/RM split too — replace aggregate COT with FM pctile for combo D/B/F.
FCI / NFCI	Runic: variable #1. SSI: NOT directly included	NFCI is in Runic but absent from SSI. SSI uses HY spreads and VIX as FCI proxies.	GAP IN SSI — NFCI should be added to SSI Layer 2 or Layer 3 as a macro confirmation signal. It is independent from the other SSI signals.
CAPE / Valuation	Runic: variable #12. SSI: NOT included	CAPE is purely a Runic Agent signal. SSI does not include valuation.	CORRECT SEPARATION — CAPE is a slow-moving structural variable unsuitable for SSI's daily/weekly signal generation. Keep in Runic only.
CNN Fear & Greed	SSI Layer 1. Runic: SSI multiplier input	CNN F&G is in SSI. The SSI score then feeds the Runic multiplier. So CNN F&G → SSI → Runic. One level removed.	CORRECT ARCHITECTURE — this is the intended flow. Not duplication, it's proper layering.
McClellan Oscillator / Breadth	SSI Layer 2. Runic: not directly included	Market breadth is SSI only. Runic does not use breadth signals.	CLEAN SEPARATION — breadth stays in SSI. The SSI output score then informs Runic via the multiplier.

Conclusion: The main overlap problem is HY spreads (measured twice, differently) and VIX term structure (measured identically). The fix: in the combined Runic + SSI system, treat HY OAS as the Runic input and HYG/LQD as the SSI input — they are the same signal, so only ONE should feed the final confidence score. VIX/VIX3M should be allocated to Runic (combo D) and EXCLUDED from SSI Layer 2, with SSI instead using a different breadth or sentiment signal in its place (McClellan is already there).

PART 9 — DIVYANSHU TEST LIST (Complete)

The following tests should be run, documented with full results (n observations, avg forward return at 1/3/6/12m, Sharpe ratio, hit rate %, worst instance), and presented back before any threshold is treated as final.

#	Test Name	What to Test	Method	Output Format

1	SSI Entry Threshold Sweep (Long)	Sweep SSI < X for X in {−0.3,−0.4,−0.5,−0.6,−0.7,−0.8,−0.9}	For each X: find all historical dates where SSI crossed below X. Measure SPX returns at 1w/2w/1m/3m/6m. Report n, avg, median, win%, worst, Sharpe.	Table + line chart of avg return vs threshold
2	SSI Entry Threshold Sweep (Short)	Sweep SSI > X for X in {+0.4,+0.5,+0.6,+0.7,+0.8,+0.9}	Same as above but for shorts. Key hypothesis: +0.8 or +0.9 produces better short signals than +0.6.	Table + line chart
3	SQUEEZE SETUP Grid Search	FM pctile 15–40 (step 5) × RM pctile 40–65 (step 5)	For each (FM,RM) pair: count instances 2006–2026. Measure avg SPX return 4wk/8wk/12wk forward. Create 6×6 heatmap.	Heatmap of forward returns
4	LIQUIDITY EXIT Grid Search	RM pctile 15–40 (step 5) × FM pctile 45–75 (step 5)	For each (RM,FM) pair: count instances. Measure avg SPX drawdown over next 4–12 weeks.	Heatmap of drawdowns
5	TP/SL Multiplier Optimisation	TP: 5–20× daily vol. SL: 8–25× daily vol	Run backtest with all TP/SL combinations on SPY 2006–2026 using long signals only. Measure Sharpe, max drawdown, win rate.	3D surface plot of Sharpe vs TP vs SL
6	CNN F&G Forward Returns	All instances of F&G < 20, < 10, > 80, > 90 from 2011–2026	Pull CNN F&G history. Mark each crossing. Measure SPY returns at 1/3/6/9/12m. Include VIX context at each instance.	Table with all instances + distribution histogram
7	DBMF Rolling Beta Threshold	21-day DBMF/SPY beta: test cutoffs −0.05, −0.10, −0.15, −0.20	For each beta threshold: identify dates where beta crossed below threshold. Measure SPY return over next 2/4/8 weeks. Run Granger causality.	Table + Granger p-value chart
8	HYG/LQD Widening Definition	4-week HYG/LQD pct change: test −1%, −1.5%, −2%, −3% as signal thresholds	For each threshold: find all crossing dates. Measure SPY return 1/4/8wks. Measure days to VIX spike after each instance.	Table + lead-time histogram
9	Z-Score vs Percentile Rank Comparison	Replace all z-score inputs with 3yr rolling percentile rank. Re-run full SSI.	Run SSI with z-scores (current) and with percentile ranks (proposed). Compare Sharpe, max drawdown, performance in 2020 and 2022 crisis periods specifically.	Side-by-side performance comparison
10	Layer 2 Confirmation Threshold	z-score > X where X in {0.0, 0.25, 0.5, 0.75, 1.0} for Layer 2 signals	For each X: count how many Layer 2 signals confirm simultaneously. Measure long signal quality (forward returns) at each confirmation threshold.	Table: threshold vs signal quality
11	VIX Regime Multiplier — Does It Help?	Test with and without the VIX regime multiplier applied to position sizing	Run full backtest 2006–2026 with multiplier active vs without. Key test: does it	Equity curve comparison. Especially

			improve Sharpe in non-crisis periods without degrading it in crisis periods?	compare Oct 2022 episode.
12	Bollinger Band + SSI Combination	Lower BB touch AND SSI < −0.6 simultaneously	Find all historical instances of lower BB touch on SPY. Subset to those where SSI was simultaneously < −0.6. Compare forward returns of the subset vs all lower BB touches.	Hit rate comparison table
13	Stochastic < 20 + McClellan Oversold	Stochastic below 20 turning up AND McClellan z-score positive simultaneously	Find all instances 2006–2026. Measure SPY returns 1/2/4wks forward. Compare to: (a) stochastic alone, (b) McClellan alone.	3-way comparison table
14	Gross/Net Divergence Revised Definition	Test: gross >75th pctile 3wks sustained + net falling + HYG/LQD < −1.0%	Identify all instances matching revised 3-condition definition. Measure SPY drawdown over next 4–12wks.	Instance list with outcomes
15	SBI Short Signal Validation	Count of short signals from BandMatrix/DeltaDrift/FractalTrack above 90th pctile of 1yr history	Find all instances. Measure SPY return 1/4/8wks. Key test: is this a useful short signal or is it noise?	Return distribution histogram

PART 10 — FRIDAY PULL LIST FOR DIVYANSHU (Every Friday, before COT report)

These are the data points that must be pulled and logged every Friday for the Runic Agent + SSI system:

Variable	Source	What to Pull	Update Timing
NFCI	FRED: NFCI	Latest weekly reading. Compute 3yr percentile. Flag if > +0.3 (RARE) or > +0.8 (EXTREME).	Thursday evening (FRED updates)
HY Credit Spreads OAS	FRED: BAMLH0A0HYM2	Latest OAS in bps. Compute 3yr pctile. Flag if > 400bps.	Daily available; pull EOD Friday
VIX Level	CBOE: ^VIX (Yahoo)	Closing VIX. Flag if > 25.	EOD Friday
VIX3M Level	Yahoo: ^VIX3M	Closing VIX3M. Compute VIX3M/VIX ratio. Flag if ratio > 1.10 (D signal) or < 0.95 (G signal).	EOD Friday
WTI Oil	Yahoo: CL=F	Friday close. Compute 4-week % change. Flag if > +10% (EXTREME) or < −10%.	EOD Friday
USD/CNH	Yahoo: USDCNH=X	Friday close. Compute 4-week % change. Flag if > ±1.5% (RARE) or ±3.5% (EXTREME).	EOD Friday
CFTC — Fast Money	CFTC.gov TFF report	Lev_Money_Positions_Long/Short_All for SPX futures. Compute 3yr pctile of net position. Flag <15th or >85th.	Released Fridays ~3:30pm ET

CFTC — Real Money	CFTC.gov TFF report	Asset_Mgr_Positions_Long/Short_All for SPX futures. Compute 3yr pctile. Flag <30th or >75th.	Released Fridays ~3:30pm ET
10Y–2Y Yield Curve	FRED: T10Y2Y	Daily spread in bps. Compute 4-week change (steepening rate). Flag if spread < −30bps or steepening > 15bps/4wk.	Daily; pull EOD Friday
Fed Balance Sheet WALCL	FRED: WALCL	Latest weekly reading in $trn. Compute MoM % change. Flag if > ±0.8% (RARE).	Thursdays H.4.1 release
Gold/Silver Ratio	FRED GOLD / SILVER or Yahoo GC=F/SI=F	Friday close of both metals. Compute GSR. Compute 4-week % change. Flag if GSR 4wk Δ > ±5% (RARE) or ±8% (EXTREME).	EOD Friday
CAPE / Shiller P/E	multpl.com	Monthly. Pull first Friday of month. Flag if > 28x (RARE) or > 32x (EXTREME).	Monthly, first Friday
CPI Surprise	BLS / Investing.com	Only on CPI release days. Compute actual vs consensus. Flag if Δ > ±0.2pp (RARE) or ±0.4pp / 2× consecutive (EXTREME).	CPI release days only
HYG/LQD Ratio	Yahoo: HYG + LQD	Daily prices. Compute HYG/LQD ratio and 4-week % change. Flag if < −1.5% (SSI Layer 2).	EOD Friday
DBMF	Yahoo: DBMF	Daily prices. Update 21-day rolling beta vs SPY. Flag if beta < −0.10.	EOD Friday
CNN Fear & Greed	money.cnn.com/data/fear-and-greed/	Current score 0–100. Flag if < 20 or > 80.	EOD Friday
AAII Bull-Bear Spread	aaii.com/sentimentsurvey	Latest weekly survey. Bulls% − Bears%. Flag if spread < −20pp or > +30pp.	Thursdays (survey releases)
NAAIM Exposure Index	naaim.org	Latest weekly reading. Flag if < 30 (extreme underweight) or > 90 (extreme overweight).	Wednesdays